

‘ilm-i ḥurūf علم حروف (letter science, lettrism)

Like kabbalah, its coeval Hebrew twin, Arabic lettrism was born of late antique Hellenistic alphanumeracy. From the eighth century onward, this basic identity of letters and numbers peculiar to the Greek, Hebrew and Arabic alphabets inspired increasingly robust theorizations of scripture—the Torah or the Quran—as mathematico-linguistic maps of history and the cosmos, and indeed the primary mechanism of Creation itself. Letter cosmogonies were accordingly developed by various early Shi‘i “extremist” (*ghulāt*) movements of West Asia and then Isma‘ili and Jewish thinkers in tandem. Lettrism and kabbalah proper, however, were both born in twelfth- and thirteenth-century Islamic and Christian Spain, whence they spread eastward, southward and northward like wildfire over the following centuries, driven by the successive apocalypses of climate change, the Reconquista, the Mongol conquest and the Black Death. In the Islamic sphere, Ibn al-‘Arabī (d. 1240) and Abū l-‘Abbās al-Būnī (d. btw. 1225-33), among other major western and eastern Sufi thinkers, came to emblemize theoretical and practical lettrism respectively. This diasporic process matured in early fifteenth-century Timurid Iran in particular, where lettrism was first systematized and reconceptualized by Ibn Turka of Isfahan (d. 1432) as the master Pythagorean, mathematizing science of Perso-Mongol Turko-Islamic universal West-East empire. (The contemporary emergence of Christian kabbalah among European Renaissance humanists, by contrast, was largely driven by insularist xenophobia.) The consciously paradigm-shifting Pythagoreanization of lettrism under the Timurids helped define the succeeding major Persianate imperial cultures (Mughal, Safavid, Ottoman and Uzbek) in turn, which by the sixteenth century spanned a full third of the human race and from the eastern Mediterranean to India; there it informed everything from science to architecture, philosophy to poetry, talismanry to calligraphy, historiography to war—all as applications of *cosmic philology*. Positing the readability and rewritability of the world, what we might call “philological consciousness” thus became a guiding scientific, technological, cultural, social and political ethos of Islamo-Afro-Eurasian early modernity, as the surviving historical, textual and material record amply attests.

Unlike kabbalah, from the Hebrew term usually translated as “reception” or “tradition,” the range of Arabo-Persian terms used to refer to lettrism emphasize the linguistic, grammatological nature of this occult science. In its medieval Arabic iterations, *‘ilm al-ḥurūf wa-l-asmā’* (“science of letters and names”) records its definitive “quranicization” under Sufi auspices, whereby quranic and extra-quranic divine names became central to lettrist practice in the form of mantras, diagrams and talismans. During the same period, the late antique term *jafr*—originally denoting the cache of secret prognosticons associated with the Holy Family and various Abrahamic prophets—was modernized by Sunni and Shi‘i scholars in tandem to marry lettrism to astrology as ultimate imperial technologies of time. In less apocalypticist but equally messianic early modern Persianate usage, both trends converged to make lettrism the matrix of divination and magic in its new role as primary vehicle of Islamic Pythagoreanism. The anarchic Sufi Ḥurūfī (Lettrist) and Nuqtavī (Dottist) movements notwithstanding, this mainstreaming is

reflected in the simplified term *‘ilm-i ḥurūf* standard in Persian from the fifteenth century onward, with *jafr* often being used metonymically for lettrism down to the present.

Most salient for the history of philosophy and science is the early modern identification of lettrism with *taḥqīq*, the free, empirical investigation of reality, marking its marriage to the mathematical sciences in the quest to riddle the source code of the cosmos, that other Book. (Saints of Science like Giordano Bruno (d. 1600), Johannes Kepler (d. 1630) and Isaac Newton (d. 1727) were committed kabbalists and/or Pythagoreans too.) By the same cosmological token, *taḥqīq* also came to denote the birth of philology itself, encoding and driving the distinctively early modern impulse to retrieve, critically edit and translate the textual record—both “Western” and “Eastern”—as perennialist yet progressivist proof of the superiority of the Moderns to the Ancients. To be a philologist was not necessarily to be a lettrist, of course; but the central lettrist, Pythagorean principle of the *coincidentia oppositorum* (*majma‘ al-aḍḍād*) or marriage of opposites served to valorize this larger comparativist, historicist and textualist ethos.

The universal prestige lettrism had acquired by the sixteenth century is signaled by its prominence in Arabic and Persian encyclopedias of the sciences, where it is classified, exceptionally, as a simultaneously natural, mathematical and metaphysical science. Many of the leading philosophers, judges, historians, litterateurs, calligraphers, mathematicians, astronomers, physicians and architects of the early modern Persian cosmopolis (sometimes the same person) accordingly either wrote lettrist works or were conversant in the science as a matter of professional course—hence the frequency with which random lettrist calculations and talisman recipes occur on manuscript flyleaves. Perhaps most emblematic of this cultural seachange was the transmogrification of Ibn Sīnā (Avicenna, d. 1037) and Ghazālī (d. 1111) themselves—no boosters of the occult sciences—into lettrists by the turn of the seventeenth century, when Mongol-era Persian grimoires written in their names became bestsellers.

Lettrism as Islamic Pythagoreanism thus came to inform, implicitly or explicitly, a wide range of Persianate disciplines, institutions and practices. In philosophy, its Pythagorean tenor helped inspire the florescence of Neoplatonism as Iamblichean theurgy that so defines early modern Illuminationist or Enlightenment (*ishrāqī*) thought—and rendered it specifically *talismanic*. In theology, the resurgence of Ash‘arī occasionalism (divine re-creation of the cosmos in every instant) appears biographically to have a lettrist link. In quranic exegesis, explications of the mysterious disconnected letter series opening 29 suras (*muqatta‘āt*)—originary inspiration and justification for lettrism itself—became more prominent and more occult-scientific in tenor, and at least one entirely lettrist tafsir was penned, in Persian, by the Timurid everyman’s polymath, preacher and occultist Ḥusayn Vā‘iz Kāshifī (d. 1502). In ritual practice, the reconceptualization of prayer as a lettrist technology led to the proliferation of manuals showing how to guarantee results mathematically. In historiography, the Timurid fusion of lettrism and astrology made the art of history an empirical science by proving time’s cyclical obedience to astro-mathematico-linguistic laws; this led to the unprecedentedly eager patronage of these twin time technologies at imperial courts, such that portable, battlefield-sized Comprehensive Prognosticons (sg. *jafr-i jāmi‘*) became the Ottoman dynastic handbook par excellence. In poetry and literary culture, the emergence of the logograph or riddle (*mu‘ammā*) genre as ubiquitous social calling card—the “poeticization of everyday life”—saw its theorization as lettrist disci-

pline, the nature or fate of the encoded person or object discoverable through name decoding—whence the ubiquity of chronograms too. In calligraphy and painting, certain leading artists, unsurprisingly, were associated with the science, and the Safavid invention of the album preface as a new genre of art history and theory was openly predicated on Ibn Turka's lettrist-Pythagorean theory of vision as information. In architecture, the rewritability of the world manifested across the Persianate world in its most grandly material imperial form: Isfahan, the new Safavid capital, was radically redesigned by the famed renaissance man and lettrist Shaykh Bahā'ī (d. 1621) as the first modern city, a perspectivist, Pythagorean machine for the production of superhumans; the Tāj Maḥal, Indian apotheosis of the Timurid imperial vision, likewise mathematically and linguistically encodes and embodies both heaven on earth and the theosis (*ta'alluh*) of humanity through the marriage of all opposites. In astronomy and mathematics, Ibn Turka's Pythagorean lettrism was philosophically, socially and politically a major factor in the mathematization of astronomy at the Samarkand Observatory now much celebrated by historians of science, emblematic in turn of the general process of *mathesis* that defines the modern "Scientific Revolution" narrative; reflecting this epochal departure from Aristotelian physics, Safavid teaching manuals feature appendices detailing the lettrist applications of even simple arithmetic, and Mughal star tables (sg. *zīj*) marry lettrism to astrology as a basic organizational and theoretical principle. In magic, Arabo-Persian grimoires—now astro-lettrist as a rule—became quite literally *grammars of being*, and bestsellers to boot; the genre represents a sizable percentage of early modern Persianate book production, whose Afro-Eurasian florescence was the realization of the lettrists' dream: *the world as text*. And *the world as number*: geomancy (*'ilm-i raml*), lettrism's sister Pythagorean, Afro-Islamic information science, and another early modern imperial mode of googling and spycraft, helped make Leibniz (d. 1716) the father of modern computing.

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